optimizing linear regression

CSCI 270

linear regression

model space

$$y = \begin{bmatrix} \theta_1 \\ \vdots \\ \theta_D \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ \vdots \\ x_D \end{bmatrix}$$

loss function

$$L(\theta) = \sum_{n=1}^{N} \left(\theta \cdot x^{(n)} - y^{(n)}\right)^2$$

optimization

$$\underset{\theta}{\operatorname{argmin}}\ L(\theta)$$

training data

instance	features				outcome
	x_1	x_2	$\cdots$	x_D	y
1	$x_1^{(1)}$	$x_2^{(1)}$	$\cdots$	$x_D^{(1)}$	$y^{(1)}$
2	$x_1^{(2)}$	$x_2^{(2)}$	$\cdots$	$x_D^{(2)}$	$y^{(2)}$
$\vdots$	$\vdots$	$\vdots$		$\vdots$	$\vdots$
N	$x_1^{(N)}$	$x_2^{(N)}$	$\cdots$	$x_D^{(N)}$	$y^{(N)}$

test data

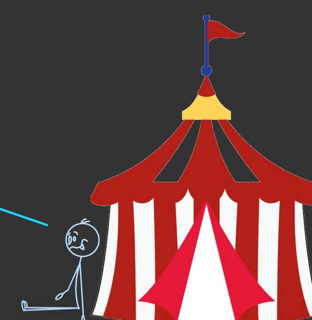
linear regression

loss function

$$L(\theta) = \sum_{n=1}^{N} \left(\theta \cdot x^{(n)} - y^{(n)}\right)^2$$

can we compute this without the explicit

loop?



loss function

$$L(\theta) = \sum_{n=1}^{N} \left(\theta \cdot x^{(n)} - y^{(n)}\right)^2$$

$$\underset{\theta}{\operatorname{argmin}} \, L(\theta)$$

$$= \underset{\theta}{\operatorname{argmin}} \sum_{n=1}^{N} \left(\theta \cdot x^{(n)} - y^{(n)}\right)^2$$

$$= \underset{\theta}{\operatorname{argmin}} \sum_{n=1}^{N} \left(\theta \cdot x^{(n)}\right)^2 - 2\left(\theta \cdot x^{(n)}\right) y^{(n)} + \left(y^{(n)}\right)^2$$

loss function

$$L(\theta) = \sum_{n=1}^{N} \left(\theta \cdot x^{(n)} - y^{(n)}\right)^2$$

$$\underset{\theta}{\operatorname{argmin}}\, L(\theta)$$

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doesn't depend on θ

loss function

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$$\underset{\theta}{\operatorname{argmin}}\, L(\theta)$$

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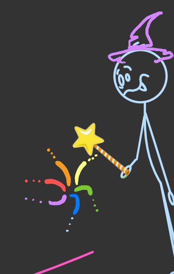


doesn't depend on θ

loss function

$$L(\theta) = \sum_{n=1}^{N} \left(\theta \cdot x^{(n)} - y^{(n)}\right)^2$$

$$\underset{\theta}{\operatorname{argmin}}\, L(\theta)$$

$$= \underset{\theta}{\operatorname{argmin}} \sum_{n=1}^{N} \left(\theta \cdot x^{(n)} - y^{(n)}\right)^2$$

$$= \underset{\theta}{\operatorname{argmin}} \boxed{\sum_{n=1}^{N} \left(\theta \cdot x^{(n)}\right)^2 - 2\left(\theta \cdot x^{(n)}\right) y^{(n)}}$$

so this loss is equivalent

loss function

$$L(\theta) = \sum_{n=1}^{N} (\theta \cdot x^{(n)})^2 - 2(\theta \cdot x^{(n)})\, y^{(n)}$$

and yet we still have the

loop

loss function

$$L(\theta) = \sum_{n=1}^{N} \left(\theta \cdot x^{(n)}\right)^2 - 2\left(\theta \cdot x^{(n)}\right) y^{(n)}$$

luckily this loss lends itself to additional magicks

$$L(\theta) = \sum_{n=1}^{N} \left[(\theta \cdot x^{(n)})^2 - 2 (\theta \cdot x^{(n)}) \, y^{(n)} \right]$$

$$= \sum_{n=1}^{N} \left[(\theta \cdot x^{(n)})^2 \right] - 2 \sum_{n=1}^{N} \left[(\theta \cdot x^{(n)}) \, y^{(n)} \right]$$

$$= \begin{bmatrix} \theta \cdot x^{(1)} \\ \vdots \\ \theta \cdot x^{(N)} \end{bmatrix} \cdot \begin{bmatrix} \theta \cdot x^{(1)} \\ \vdots \\ \theta \cdot x^{(N)} \end{bmatrix} - 2 \begin{bmatrix} \theta \cdot x^{(1)} \\ \vdots \\ \theta \cdot x^{(N)} \end{bmatrix} \cdot \begin{bmatrix} y^{(1)} \\ \vdots \\ y^{(N)} \end{bmatrix}$$

$$= \theta^\top X^\top X \theta - 2 \, \theta^\top X^\top y$$

$$L(\theta) = \sum_{n=1}^{N} \left[\left(\theta \cdot x^{(n)}\right)^2 - 2\left(\theta \cdot x^{(n)}\right) y^{(n)} \right]$$

$$= \sum_{n=1}^{N} \left[\left(\theta \cdot x^{(n)}\right)^2 \right] - 2 \sum_{n=1}^{N} \left[\left(\theta \cdot x^{(n)}\right) y^{(n)} \right]$$

$$= \begin{bmatrix} \theta \cdot x^{(1)} \\ \vdots \\ \theta \cdot x^{(N)} \end{bmatrix} \cdot \begin{bmatrix} \theta \cdot x^{(1)} \\ \vdots \\ \theta \cdot x^{(N)} \end{bmatrix} - 2 \begin{bmatrix} \theta \cdot x^{(1)} \\ \vdots \\ \theta \cdot x^{(N)} \end{bmatrix} \cdot \begin{bmatrix} y^{(1)} \\ \vdots \\ y^{(N)} \end{bmatrix}$$

$$= \theta^{T} X^{T} X \theta - 2\, \theta^{T} X^{T} y$$

$$\begin{bmatrix} \theta\cdot x^{(1)} \\ \vdots \\ \theta\cdot x^{(N)} \end{bmatrix} \cdot \begin{bmatrix} \theta\cdot x^{(1)} \\ \vdots \\ \theta\cdot x^{(N)} \end{bmatrix}$$

$$= \begin{bmatrix} \theta\cdot x^{(1)} & \cdots & \theta\cdot x^{(N)} \end{bmatrix} \begin{bmatrix} \theta\cdot x^{(1)} \\ \vdots \\ \theta\cdot x^{(N)} \end{bmatrix}$$

$$= \begin{bmatrix} \theta \end{bmatrix} \begin{bmatrix} x^{(1)} & \cdots & x^{(N)} \end{bmatrix} \begin{bmatrix} x^{(1)} \\ \vdots \\ x^{(N)} \end{bmatrix} \begin{bmatrix} \theta \end{bmatrix}$$

$$= \underset{\begin{bmatrix} \theta_1 & \cdots & \theta_D \end{bmatrix}}{\theta^T} \; \underset{\begin{bmatrix} x_1^{(1)} & \cdots & x_1^{(N)} \\ \vdots & & \vdots \\ x_D^{(1)} & \cdots & x_D^{(N)} \end{bmatrix}}{X^T} \; \underset{\begin{bmatrix} x_1^{(1)} & \cdots & x_D^{(1)} \\ \vdots & & \vdots \\ x_1^{(N)} & \cdots & x_D^{(N)} \end{bmatrix}}{X} \; \underset{\begin{bmatrix} \theta_1 \\ \vdots \\ \theta_D \end{bmatrix}}{\theta}$$

$$= \boxed{\begin{bmatrix} \theta\cdot x^{(1)} \\ \vdots \\ \theta\cdot x^{(N)} \end{bmatrix} \cdot \begin{bmatrix} \theta\cdot x^{(1)} \\ \vdots \\ \theta\cdot x^{(N)} \end{bmatrix}} - 2 \begin{bmatrix} \theta\cdot x^{(1)} \\ \vdots \\ \theta\cdot x^{(N)} \end{bmatrix} \cdot \begin{bmatrix} y^{(1)} \\ \vdots \\ y^{(N)} \end{bmatrix}$$

$$= \boxed{\theta^T X^T X \theta} - 2\, \theta^T X^T y$$

$$2\begin{bmatrix} \theta \cdot x^{(1)} \\ \vdots \\ \theta \cdot x^{(N)} \end{bmatrix} \cdot \begin{bmatrix} y^{(1)} \\ \vdots \\ y^{(N)} \end{bmatrix}$$

$$\begin{aligned} &= \begin{bmatrix} \theta \cdot x^{(1)} \\ \vdots \\ \theta \cdot x^{(N)} \end{bmatrix} \cdot \begin{bmatrix} \theta \cdot x^{(1)} \\ \vdots \\ \theta \cdot x^{(N)} \end{bmatrix} - 2\begin{bmatrix} \theta \cdot x^{(1)} \\ \vdots \\ \theta \cdot x^{(N)} \end{bmatrix} \cdot \begin{bmatrix} y^{(1)} \\ \vdots \\ y^{(N)} \end{bmatrix} \\ &= \theta^{\top} X^{\top} X \theta - 2\,\theta^{\top} X^{\top} y \end{aligned}$$

$$= 2 \begin{bmatrix} \theta_1 & \cdots & \theta_D \end{bmatrix} \begin{bmatrix} x_1^{(1)} & \cdots & x_1^{(N)} \\ \vdots & & \vdots \\ x_D^{(1)} & \cdots & x_D^{(N)} \end{bmatrix} \begin{bmatrix} y^{(1)} \\ \vdots \\ y^{(N)} \end{bmatrix} = 2\theta^{\top} X^{\top} y$$

our loop has been transmogrified

into tensor operations

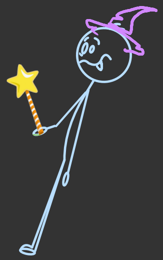
loss function

$$L(\theta) = \sum_{n=1}^{N} \left(\theta \cdot x^{(n)} - y^{(n)}\right)^2$$

$\rightarrow$

loss function

$$L(\theta) = \theta^T X^T X \theta - 2\theta^T X^T y$$



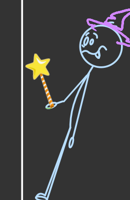
linear regression

model space

$$y = \begin{bmatrix} \theta_1 \\ \vdots \\ \theta_D \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ \vdots \\ x_D \end{bmatrix}$$

loss function

$$L(\theta) = \theta^T X^T X \theta - 2\theta^T X^T y$$



training data

	features				outcome
instance	x_1	x_2	$\cdots$	x_D	y
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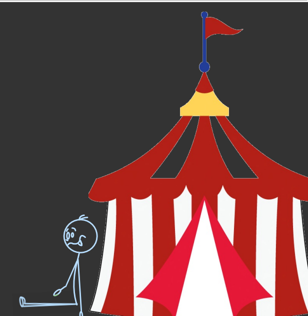
optimization

$$\underset{\theta}{\operatorname{argmin}}\, L(\theta)$$

test data



linear regression

model space

$$y = \begin{bmatrix}\theta_1 \\ \vdots \\ \theta_D\end{bmatrix} \cdot \begin{bmatrix}x_1 \\ \vdots \\ x_D\end{bmatrix}$$

training data

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loss function

$$L(\theta) = \theta^T X^T X \theta - 2\theta^T X^T y$$

optimization

$$\underset{\theta}{\operatorname{argmin}}\ L(\theta)$$

but to optimize the loss,
we need its gradient

if $L(\theta) = \theta^T X^T X \theta - 2\theta^T X^T y$, what is ∇L?

$$\nabla L = \frac{d}{d\theta}\left(\theta^T X^T X \theta\right) - \frac{d}{d\theta}\left(2\theta^T X^T y\right)$$

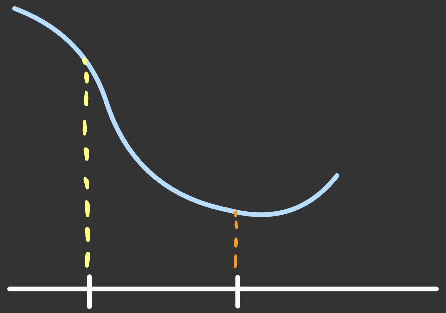
$$\boxed{\frac{d}{da}\left(a^T Z a\right) = \left(Z + Z^T\right)a} \qquad \boxed{\frac{d}{da}\left(a^T b\right) = b}$$

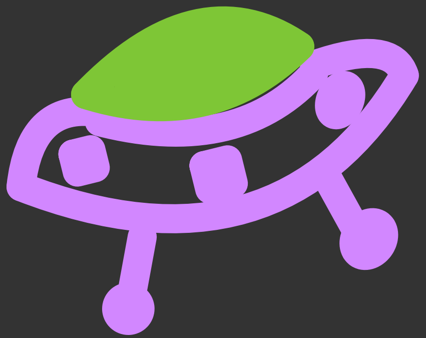
$$= \left(X^T X + \left(X^T X\right)^T\right)\theta - 2X^T y$$

$$= 2X^T X \theta - 2X^T y$$

so to minimize loss

$$L(\theta) = \theta^T X^T X \theta - 2\theta^T X^T y$$





we can run gradient

descent using

$$\nabla L = 2X^T X\theta - 2X^T y$$

but in this case
gradient descent is

overkill

a more basic approach works

solve $\frac{dL}{d\theta} = 0$

$$2X^TX\theta - 2X^Ty = 0$$

$$\Rightarrow \quad 2X^TX\theta = 2X^Ty$$

$$\Rightarrow \quad X^TX\theta = X^Ty$$

$$\Rightarrow \quad \theta = (X^TX)^{-1}X^Ty$$

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loss function

$$L(\theta) = \theta^T X^T X \theta - 2\theta^T X^T y$$

optimization

$$\underset{\theta}{\operatorname{argmin}}\, L(\theta) = (X^T X)^{-1} X^T y$$

training data

	features				outcome
instance	x_1	x_2	$\cdots$	x_D	y
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test data